

# Reflection and midpoint rigidity

[Back to Project mathematical routes](#)

## Scope

A bounded symmetric lens and reflection-center rigidity identify midpoints under an isometry. Containment in matching ambient balls supplies the local hypotheses for midpoint preservation.

5 direct Cards + 0 reused prerequisites = 5 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

## Cards in this route

- Key result: [Midpoint preservation on a quarter ball](#)
- Key definition: [Symmetric lens](#)

[Read this route with prerequisites](#)

## Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#)

## Level 0 (2 Cards)

Level 0

### Symmetric lens

Names the intersection of two closed balls with a common radius.

`MathlibAnnex.symmetricLens`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Midpoint preservation on a symmetric lens](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

### Center transport under bounded point-reflection symmetry

Uses a doubling argument to identify the reflection centers under a bijective isometry.

`MathlibAnnex.IsometryEquiv.map_center_of_mapsTo_pointReflection`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Midpoint preservation on a symmetric lens](#)

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## Level 1 (1 Card)

Level 1

### Midpoint preservation on a symmetric lens

Identifies the centers of two lenses without prescribing images of their foci.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens`

Immediate Card prerequisites: [Center transport under bounded point-reflection symmetry](#) · [Symmetric lens](#)

Used by in this scope: [Midpoint preservation under lens containment](#)

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## Level 2 (1 Card)

Level 2

### Midpoint preservation under lens containment

Restricts a set isometry and its inverse to the two contained lenses.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens_subset`

Immediate Card prerequisites: [Midpoint preservation on a symmetric lens](#)

Used by in this scope: [Midpoint preservation on a quarter ball](#)

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## Level 3 (1 Card)

Level 3

### Midpoint preservation on a quarter ball

Supplies both lens inclusions from matching ambient balls.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_mem_ball`

Immediate Card prerequisites: [Midpoint preservation under lens containment](#)

Used by in this scope: None in this selected scope

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